1.5 Scope of the Project

The scope of the Web-Based Hotel Management System for Dutch Point Negombo defines the functional boundaries, operational environment, and constraints governing the project's development. This section outlines the specific in-scope and out-of-scope boundaries, contrasts the solution against existing industrial hospitality platforms, and provides rigorous strategic evaluations through SWOT and PESTLE analyses.

1.5.1 Project Scope & System Boundaries

A. In-Scope Features (Implemented for the System):

- **User Authentication & Role-Based Access Control (RBAC):** Provision of distinct interfaces and permission boundaries for Administrators, Receptionists, and Customers to safeguard system integrity.
- **Room Inventory Management:** Centralized database management of rooms and luxury suites (Standard Queen, Deluxe Double, Ocean View Suite, Executive Suite, Presidential Penthouse, Family Villa) with dynamic pricing, amenities, capacities, and visual asset linking.
- **Real-Time Availability Engine:** Dynamic query engine that evaluates live date ranges and highlights room availability without requiring full-page reloads.
- **Automated Double-Booking Prevention:** Strict transactional validation logic in the backend enforcing non-overlapping date rules (CheckIn < RequestedCheckOut AND CheckOut > RequestedCheckIn), immediately returning conflict alerts when an overlap occurs.
- **Occupied Rooms Front-Desk Tracker:** Interactive operational dashboard allowing hotel staff to monitor all currently occupied suites, inspect guest details (name, phone, email), verify stay dates, and execute check-in/out updates.
- **Customer Self-Service Reservation Portal:** Online guest reservation workflow featuring live price calculation based on stay duration, special requests submission, booking history tracking, and direct reservation cancellation.

B. Out-of-Scope (Future Enhancements Beyond Interim Scope):

- **Live Banking Payment Gateway Integration:** Real-time payment gateway processing (e.g., Stripe, PayHere, Commercial Bank IPG) is simulated via calculated billing records for the interim evaluation phase.
- **Third-Party Channel Manager Synchronization:** Two-way XML/iCal integration with external Online Travel Agencies (OTAs) such as Booking.com, Agoda, and Expedia.
- **Smart Keycard / IoT Hardware Lock Integration:** Automated RFID or digital room lock synchronization.
- **AI-Based Dynamic Surge Pricing:** Algorithmic pricing adjustments based on real-time seasonal flight volumes or weather predictions.

1.5.2 Comparison with Existing Industrial Systems

To establish the industrial relevance of the proposed system, it is evaluated against the traditional manual methods predominantly used in smaller hospitality establishments, as well as complex commercial enterprise solutions (e.g., Oracle Opera, Cloudbeds):

Dimension	Traditional / Manual System (Logbooks & Sheets)	Commercial Enterprise PMS (e.g., Oracle Opera, Cloudbeds)	Proposed System (Dutch Point Negombo PMS)
Double-Booking Risk	High: Frequent scheduling conflicts and human errors during phone bookings.	Low: Prevented via heavy enterprise database locking.	Zero: Strictly prevented via automated transactional overlap validation.
Operational Cost	Low initial setup, but high financial loss from booking disputes and overhead.	Very High: $150 to $600 per month in recurring fees plus onboarding training costs.	Cost-Effective: Zero recurring license fees, open-source stack (React, .NET, MySQL).
System Complexity	None, but completely unscalable and vulnerable to physical record loss.	High Complexity: Bloated enterprise menus requiring specialized staff training.	Intuitive & Focused: Clean boutique resort interface with zero operational clutter.
Guest Direct Booking	None: Requires telephone calls, social media messages, or manual emails.	Supported via complex, external embedded booking widgets.	Fully Integrated: Seamless, native direct online reservation with instant feedback.
Real-Time Occupancy	Manual check of physical ledgers or Excel sheets.	Available, but buried across deep enterprise administrative submenus.	Instant Visibility: Dedicated 1-click Occupied Rooms tracker with guest records.

1.5.3 SWOT Analysis

A SWOT analysis was conducted to assess the internal technical strengths and weaknesses of the proposed system, alongside the external market opportunities and threats within the boutique hospitality sector:

• STRENGTHS (Internal)

- Automated Double-Booking Prevention algorithm ensuring data integrity.
- High-performance, lightweight 3-tier architecture (React 19 + ASP.NET Core 10 + MySQL 8.0).
- Role-Based Access Control ensuring secure operational separation for staff and guests.

• WEAKNESSES (Internal)

- Interim evaluation uses plain-text authentication (to be upgraded to JWT + salted hashing).
- Absence of direct OTA channel management (e.g., Booking.com synchronization).
- Payment handling is currently record-based and lacks live merchant gateway settlement.

- Boutique-tailored responsive UI requiring minimal staff onboarding.

• OPPORTUNITIES (External)

- Substantial reduction in third-party OTA commissions (saving 15% to 25% per booking).
- Integration with Sri Lankan online payment gateways (e.g., PayHere, Sampath/Commercial Bank IPG).
- Expansion of the system to manage beachfront dining, cabana bookings, and restaurant POS.

• THREATS (External)

- Cybersecurity threats and denial-of-service risks if deployed without production SSL/WAF.
- Internet connectivity instability in coastal regions disrupting real-time front-desk operations.
- Rapid evolution of competing cloud-based micro-SaaS hospitality tools.

1.5.4 PESTLE Analysis

- **Political (P):** The Sri Lankan Ministry of Tourism and the Sri Lanka Tourism Development Authority (SLTDA) actively support digital modernization within tourist establishments to enhance international visitor experiences. The software must adhere to municipal guest registration standards for foreign tourists.
- **Economic (E):** Boutique hotels typically lose significant revenue to 15% to 20% OTA commission rates. By offering direct online reservations, the hotel maximizes profit margins while avoiding costly enterprise PMS subscriptions, providing high economic feasibility for small-to-medium coastal resorts.
- **Social (S):** Travelers increasingly expect seamless online reservation capabilities from mobile devices. Clear room visuals, transparent rates, and instant digital booking confirmation significantly improve guest satisfaction, trust, and brand perception.
- **Technological (T):** Leveraging ASP.NET Core 10 Web API and React ensures cross-device responsiveness, high throughput, and clean RESTful decoupling. Storing data in MySQL 8.0 guarantees ACID transactional integrity, essential for robust booking concurrency and conflict prevention.
- **Legal (L):** Customer data (names, email addresses, contact numbers) must be stored and processed lawfully in strict compliance with Sri Lanka's Personal Data Protection Act (PDPA) No. 9 of 2022 and international data privacy norms (GDPR principles). Transparent cancellation terms protect both the business and consumer.
- **Environmental (E):** Transitioning from physical logbooks, paper vouchers, and printed registration cards to a centralized digital database eliminates recurring paper consumption, supporting sustainable eco-friendly tourism practices.

Summary: By clearly outlining the system boundaries, industrial benchmarks, SWOT, and PESTLE factors, this scope establishes the operational feasibility of the Dutch Point Negombo PMS. Section 1.6 will detail the tangible deliverables produced across the project lifecycle.